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Project Name: Jet Cultivation

Experiment 8: Architectural Diagram

MVC Architecture – Smart Agriculture Management System (Jet Cultivation Project)

Aim

To design and represent the MVC (Model–View–Controller) Architecture for the Smart Agriculture Management System to organize the application into structured components for better development and maintenance.

Procedure:

Objective

- To separate application logic into Model, View, and Controller
- To improve scalability and maintainability
- To handle user interaction efficiently

System Overview

The system is designed for farmers and administrators to manage agricultural activities like:

- Soil analysis
- River monitoring
- Crop management
- Selling center tracking

The architecture follows the MVC design pattern, where each component has a specific role.

Architecture Components

1. User (Farmer)

- Acts as the primary actor
- Interacts with the system through UI
- Sends requests like:
 - Login
 - View dashboard
 - Check river status
 - Analyze soil
 - View selling centers

View Layer

The View represents the user interface.

Includes:

- Login / Register Page
- Dashboard
- River Status UI
- Soil Analysis UI
- Crop Journey UI
- Centers UI

It displays data to the user and sends user actions to the controller.

Controller Layer

The Controller acts as an intermediary between View and Model.

Includes:

- Auth Controller
- Dashboard Controller
- Irrigation Controller
- Soil Controller
- Crop Controller
- Center Controller

Responsibilities:

- Receives user requests
- Processes input
- Calls appropriate model
- Returns response to View

Model Layer

The Model handles data and business logic.

Includes:

- Farmer Model
- River Model
- Soil Model
- Crop Model
- Center Model

Responsibilities:

- Data processing
- Business logic implementation
- Interaction with database

Database Layer

Stores all application data.

Tables:

- Farmers Table
- River Data
- Soil Data
- Crop Data
- Centers Data

Database Server

- Executes database queries
- Stores and retrieves data
- Communicates with Model using DB Queries

- Working Flow of the System
 1. Farmer interacts with View (UI)
 2. View sends user request to Controller
 3. Controller processes request
 4. Controller calls Model
 5. Model interacts with Database Server
 6. Data is fetched/stored
 7. Response is sent back:
 - Model → Controller → View → User

This ensures separation of concerns

Advantages of MVC Architecture

- Easy to maintain
- Reusable components
- Better code organization
- Supports multiple views
- Scalable for future development

Result

The MVC Architecture Diagram for the Smart Agriculture Management System is successfully designed, clearly showing the interaction between User, View, Controller, Model, and Database.

